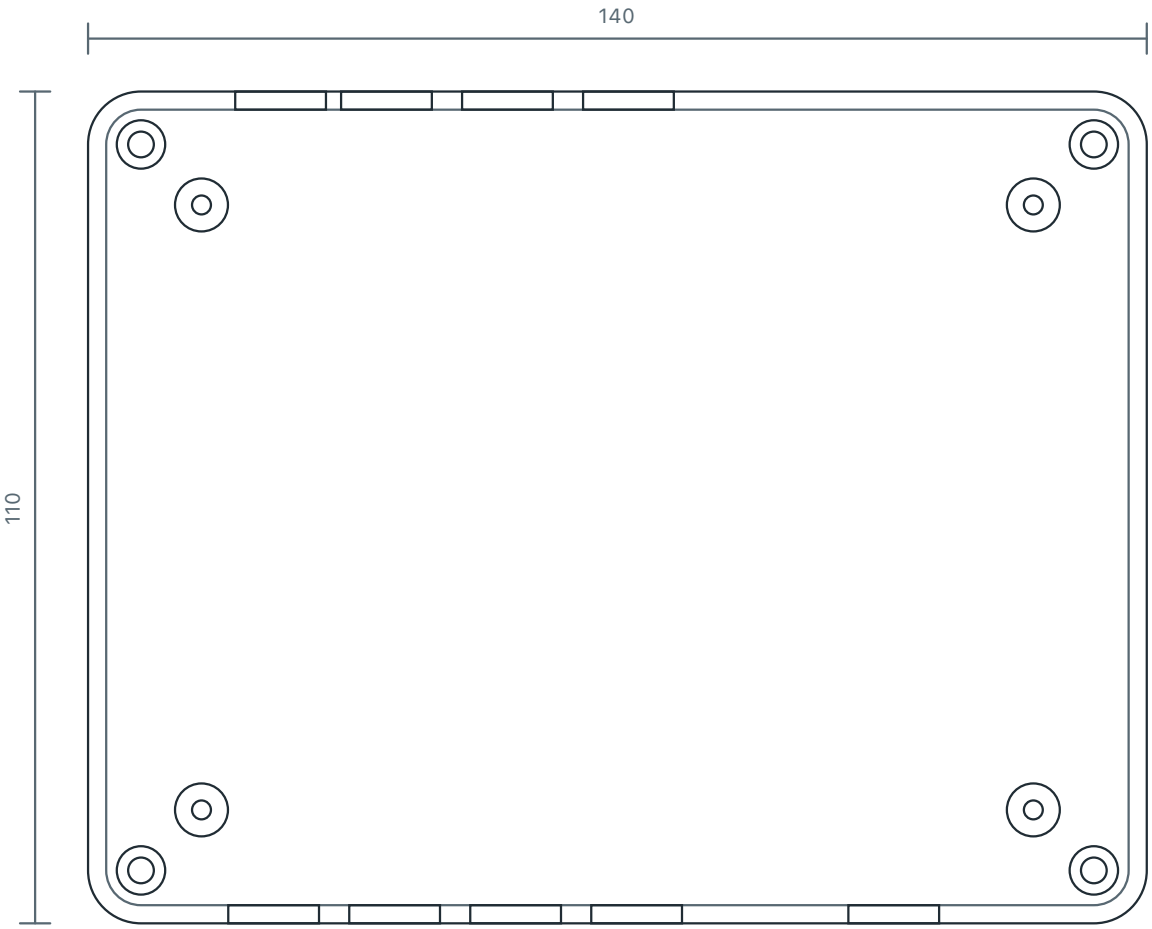
01 Base and concealed fastening



Base H 34; floor/wall 2.4. Corner R7. PCB underside Z8; PCB thickness 1.6.

PCB posts: Ø7 / pilot Ø2.5. Centers (15,15), (125,15), (15,95), (125,95). M3 × 6 + 0.5 washer.

Lid screws enter from below: M3 × 10. Centers (7,7), (133,7), (7,103), (133,103). Ø3.4 clearance; Ø6.4 × 1.5 counterbore.

Ports: 12 W × 16 H; bottom Z9. Right USB port: Y43.5–62.5, Z7–17. Switch detail: sheet 04.

Plan uses the numeric board/case XY datum. DI edge Y=0; supply/Q edge Y=110. All tolerances are prototype starting assumptions.

100 mm — verify printed scale

02 Lid, optics and display mounting

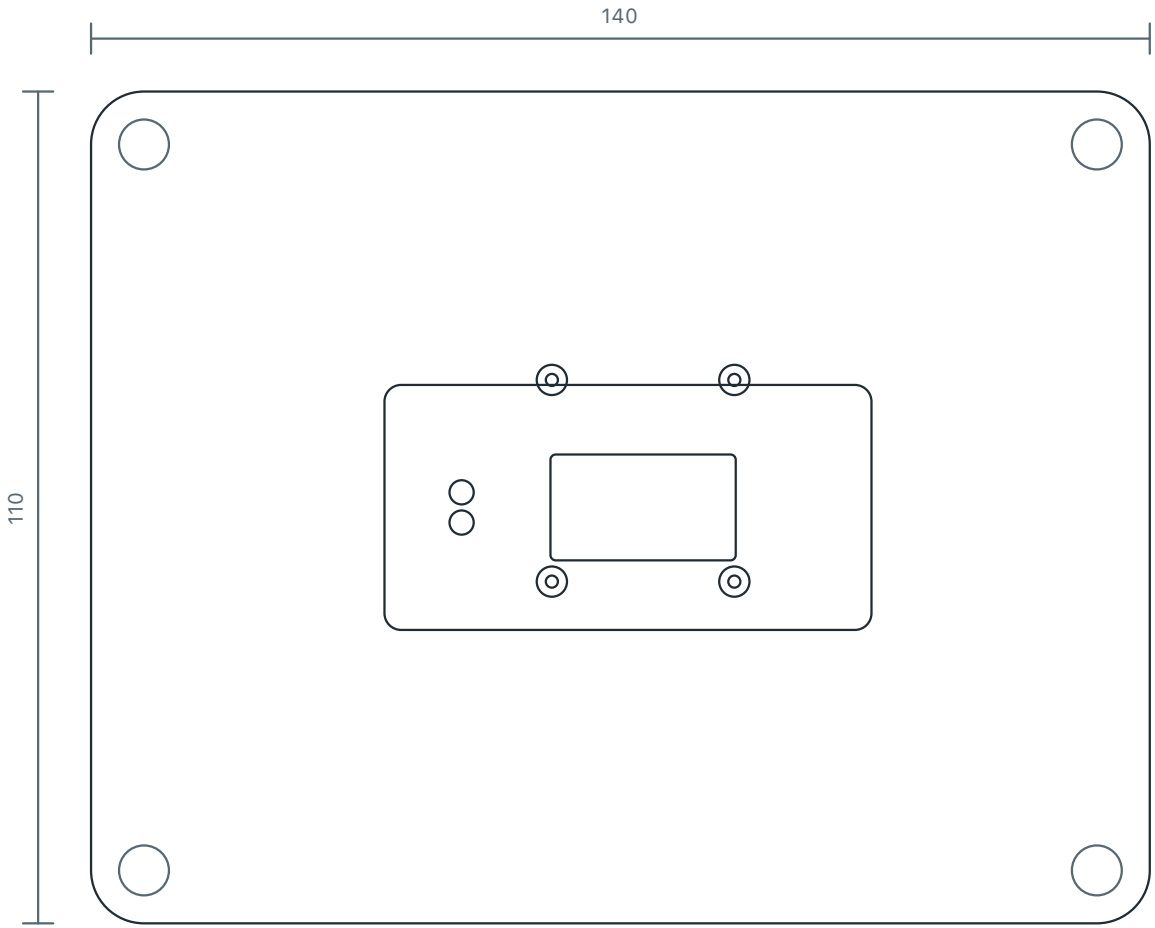


Plate Z34–37; top-edge fillet R0.7. Locating lip Z32.2–34; gap 0.3/side.

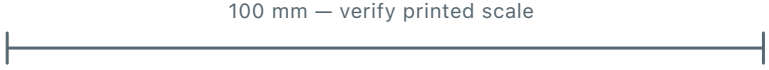
Lid columns Ø6.6, Z4.3–34. Blind M3 pilots Ø2.5, 10 deep from underside.

Lens 64 × 32 × 1, R2. Pocket 64.4 × 32.4 × 1.2; 0.2 adhesive allowance.

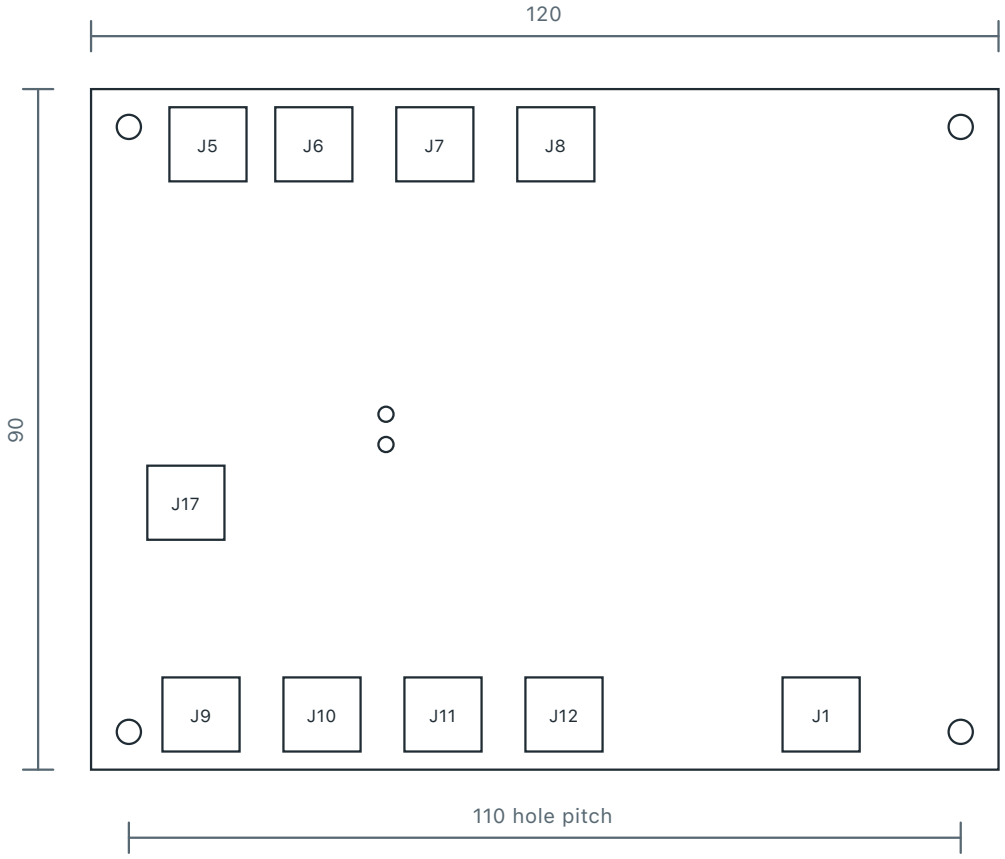
OLED PCB origin (58.395,35.599,27). Hole pitch 24.13 × 26.67. Posts Ø4; M2 pilots Ø1.6.

Glass top Z33.2; nominal lid gap 0.8. Rods Ø3 × 24.6, Z11.2–35.8. Physical fit must be checked.

OLED and column circles are underside features; no top screw holes. Top lettering is applied separately. Rods must clear LED packages.



03 Custom PCB card fit template



120 × 90 × 1.6 mm custom controller. Four Ø3.2 NPTH holes at (5,5), (115,5), (5,85), (115,85).

Case offset: +10,+10. DI edge is Y=0. Q and input supply edge is Y=90. J1.1 = +24 V; J1.2 = 0 V/GND (–).

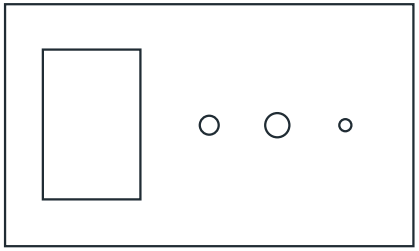
J17 is the internal switch loop. Both contacts carry positive supply; neither is ground. D2 RUN at (39,43); D3 PWR at (39,47).

Cut from card for fit checks. This is not a copper/pad image. A1 has no routed tracks and is not ready for fabrication.

100 mm — verify printed scale

04 Switch coupon, lens and optical rods

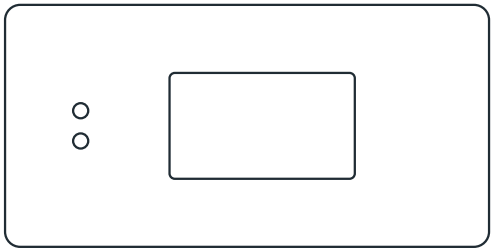
COUPON 54 × 32 × 2.4



Switch opening 12.90 × 19.81

Pilots Ø2.5 / rod bore Ø3.2 / pilot Ø1.6

LENS CUT 64 × 32 × 1 / R2



Clear screen + LED windows in a reverse black mask.

Switch panel: 2.4 mm. Opening centered at left-face Y49, Z22. C&K D102 S2 nominal dimensions; validate with actual switch.

Lens front-view mask: screen window centered at (34,16), 24.5 × 14. LED clear spots Ø2 at (10,14) PWR and (10,18) RUN.

Two clear acrylic rods: Ø3 × 24.6. Polish ends. Lower ledge Z11.2; LED top envelope Z10.6. No contact pressure on LEDs.

Base print: floor down. Lid print: outside down, columns up. Use the STL orientations; import units as mm.

100 mm — verify printed scale